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3D CAD Editing With VLMs

Autodesk Problem Overview

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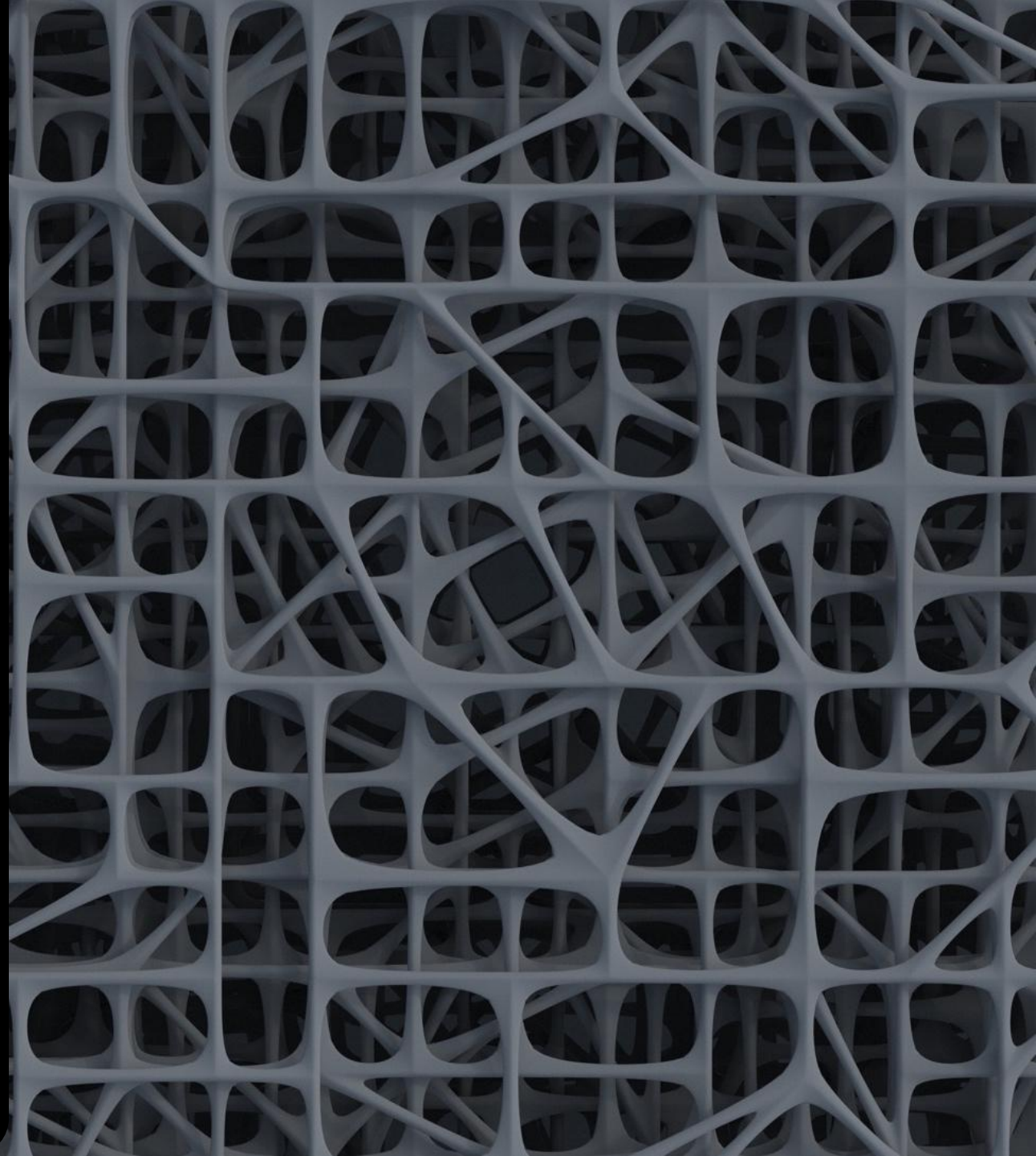
Agenda

01 The Problem

02 Your Goal

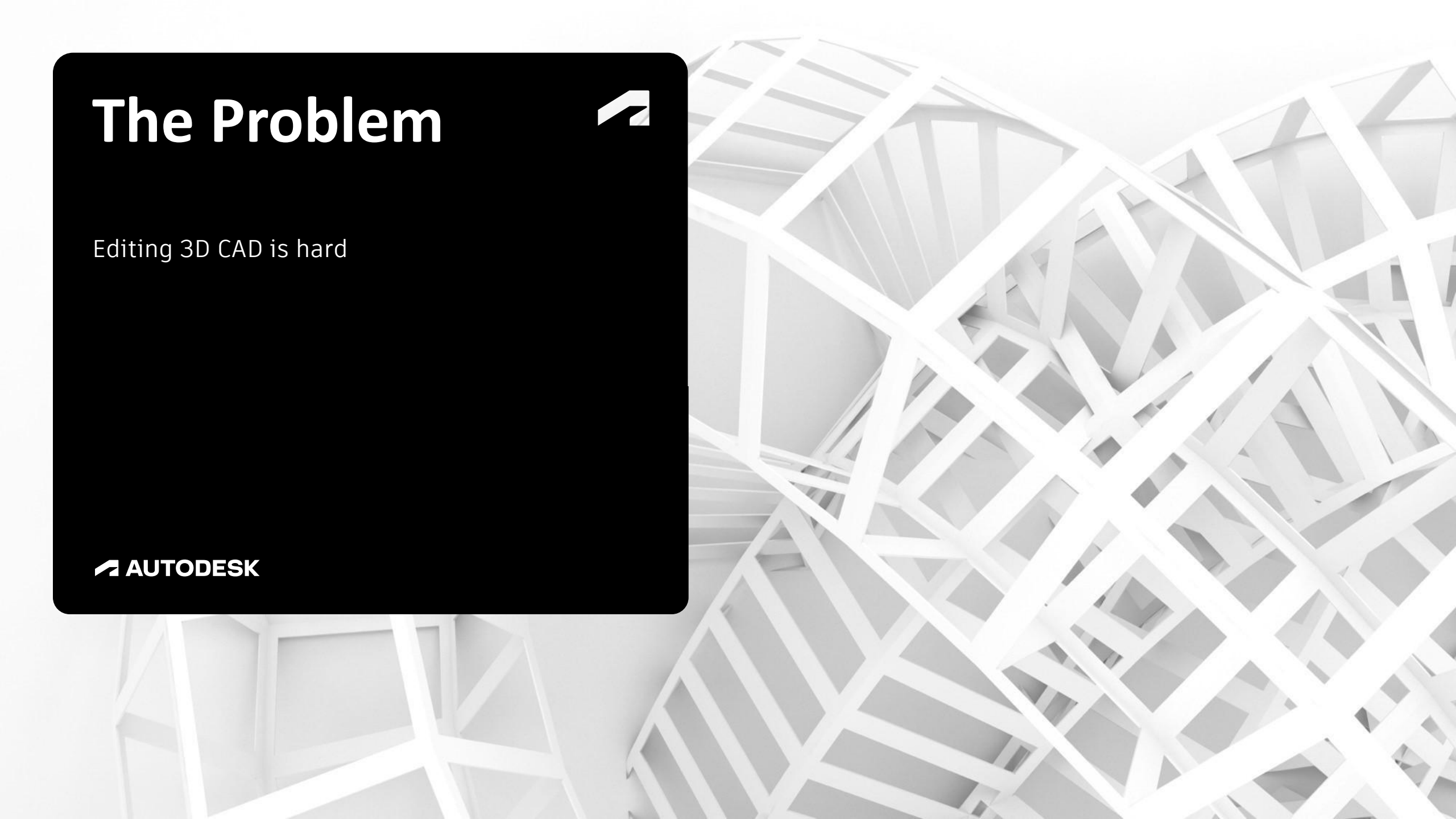
03 Overview of the Benchmark

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The Problem

Editing 3D CAD is hard

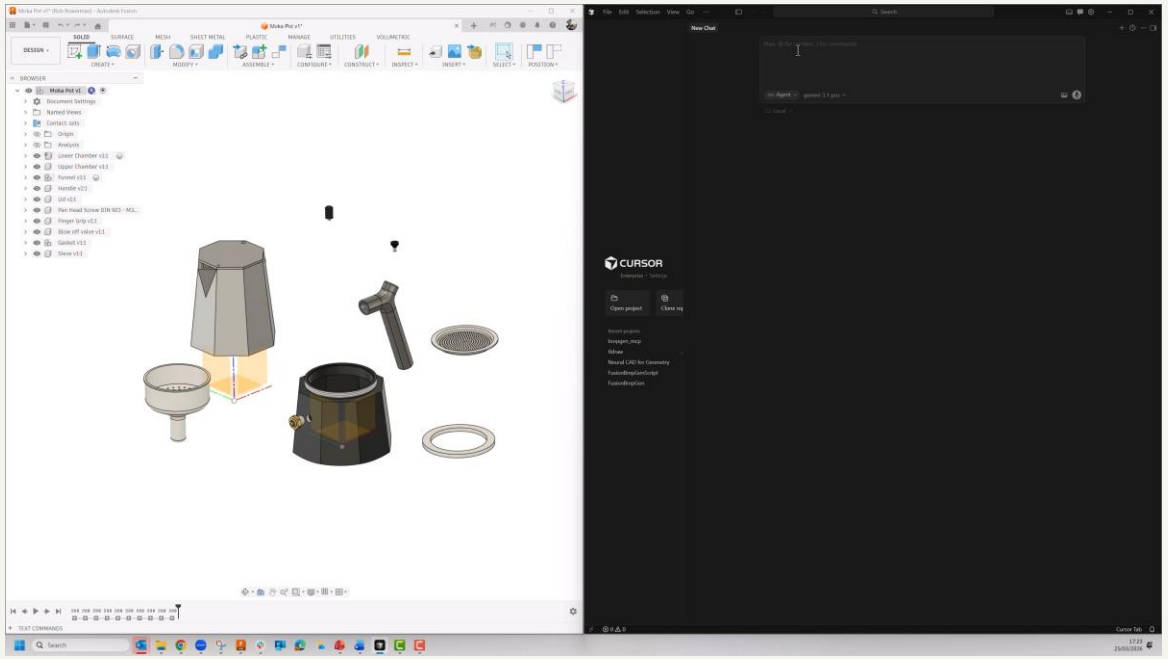


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3D CAD *generation* has progressed rapidly



Autodesk's neural CAD models



VLMs/ MCP servers

Designers spend more time *editing* than creating from scratch

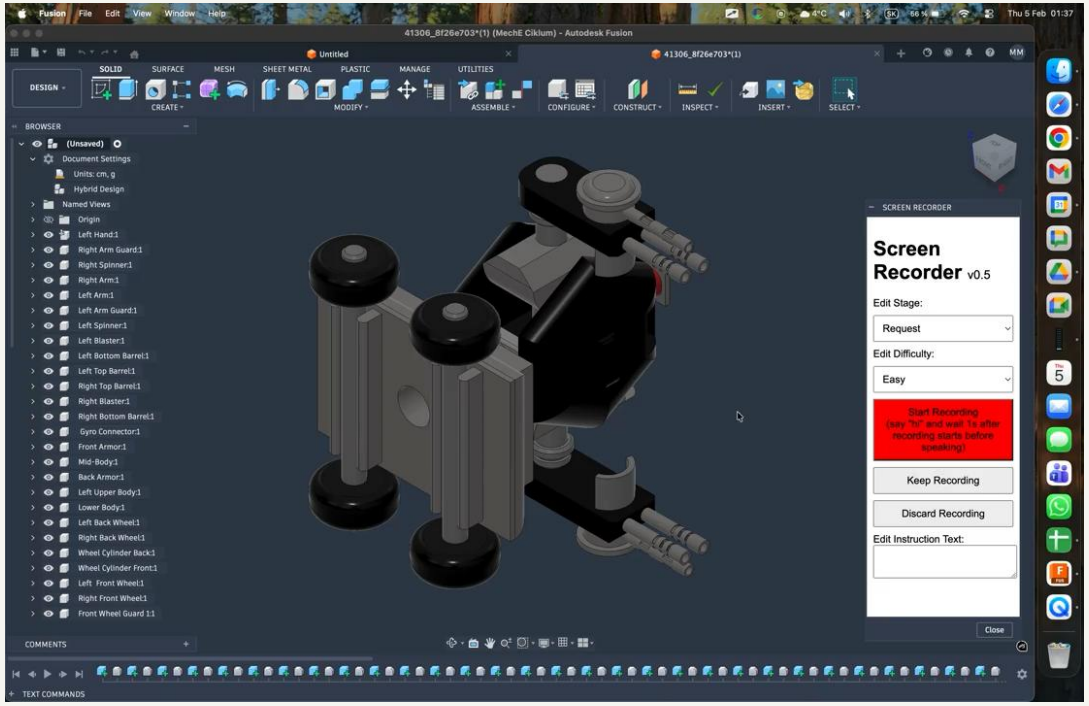
Request ID: B7A2N74ZJBF9MZHU_1770251861.898911
Modality: draw-temp
Difficulty: easy
Edit ID: DYP2J9CYL7X46RLQ_1770358371.309453
Edit Duration: 1m 47s



Repeat Measure

Editing is behind because of *data*





Datasets contain *completed CAD models* rather than *edits (differences)*

Intent of edits isn't captured

neuralCAD-Edit

An Expert Benchmark for Multimodal-Instructed 3D CAD Model Editing

neuralCAD-Edit: An Expert Benchmark for Multimodal-Instructed 3D CAD Model Editing

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Abstract. We introduce neuralCAD-Edit, the first benchmark for editing 3D CAD models collected from expert CAD engineers. Instead of text conditioning as in prior works, we collect realistic CAD editing requests by capturing videos of professional designers, interacting directly with CAD models in CAD software, while talking, pointing and drawing. We recruited ten consenting designers to contribute to this contained study. We benchmark leading foundation models against human CAD experts carrying out edits, and find a large performance gap in both automatic metrics and human evaluations. Even the best foundation model (GPT 5.2) scores 53% lower (absolute) than CAD experts in human acceptance trials, demonstrating the challenge of neuralCAD-Edit. We hope neuralCAD-Edit will provide a solid foundation against which 3D CAD editing approaches and foundation models can be developed. Code/data: https://autodeskailab.github.io/neuralCAD-Edit

Keywords: 3D CAD · 3D Editing · Multimodal · Benchmark

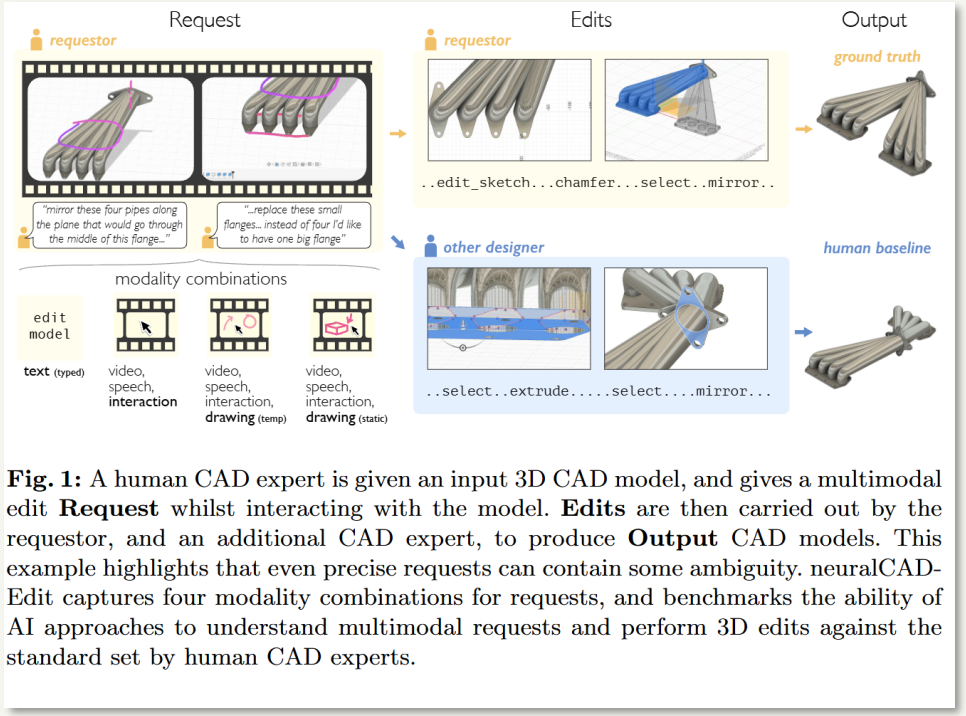
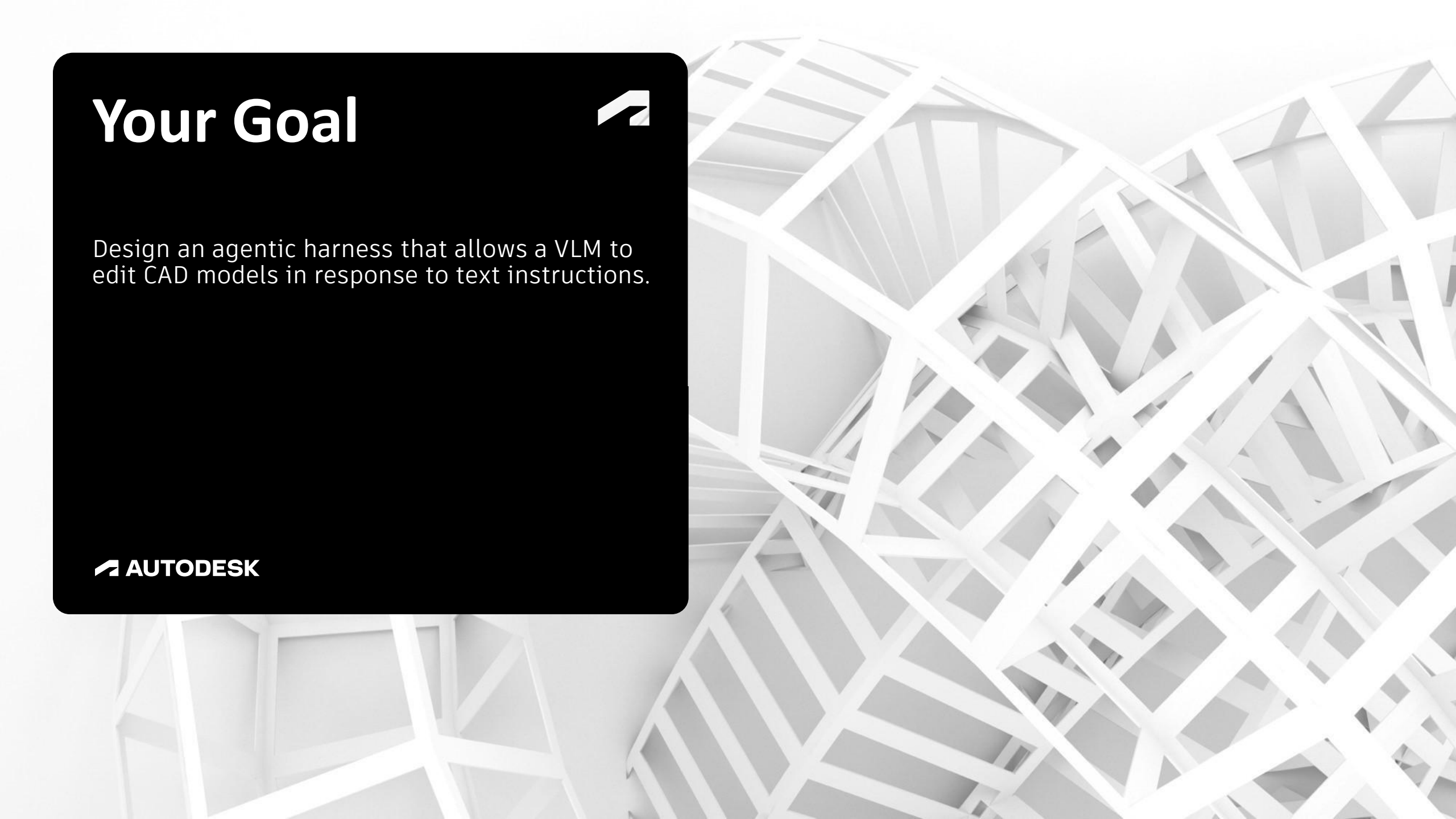


Fig. 1: A human CAD expert is given an input 3D CAD model, and gives a multimodal edit **Request** whilst interacting with the model. **Edits** are then carried out by the requestor, and an additional CAD expert, to produce **Output** CAD models. This example highlights that even precise requests can contain some ambiguity. neuralCAD-Edit captures four modality combinations for requests, and benchmarks the ability of AI approaches to understand multimodal requests and perform 3D edits against the standard set by human CAD experts.

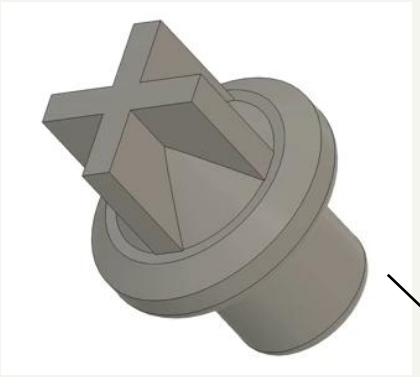
Your Goal

Design an agentic harness that allows a VLM to edit CAD models in response to text instructions.

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A Simple Harness



.step

Add a connecting hole of 1.7 mm diameter and apply 0.1 mm grooves to increase grip and prevent slipping.

text

Harness

VLM

```
# ------------------------------
# 2) Fillet sharp edges for safe handling
# ------------------------------
# Avoid global fillet; iteratively fillet the outer edges
first.
cur_shape = wp.val() if hasattr(wp, "val") else wp

edge_pts = []
try:
    for e in cur_shape.Edges():
        mp = e.Center()
        mpv = cq.Vector(mpv_x := mp.x, mpv_y := mp.y, mpv_z :=
        mp.z)

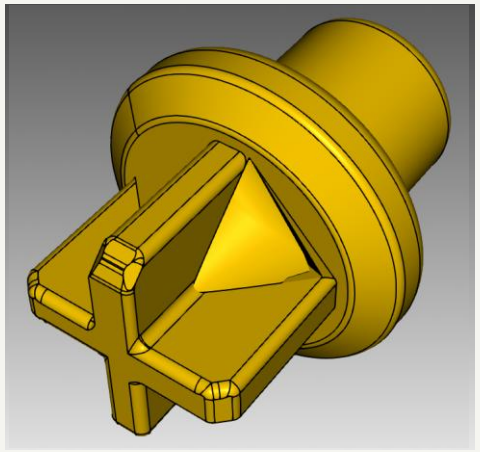
        # prioritize outer edges using transverse radius
        uvec = (mpv - c) - axis_vec.multiply((mpv - c).dot
        (axis_vec))
        r = uvec.Length

        try:
            L = e.Length()
        except Exception:
            L = None
        if L is not None and L < 2.0 * fillet_r:
            continue

        try:
            gt = e.geomType()
        except Exception:
            gt = None
        sharp_bonus = 1.0 if gt == "LINE" else 0.0

        score = r + 0.20 * sharp_bonus + 0.01 * (L or 0.0)
        edge_pts.append((score, cq.Vector(mpv_x, mpv_y,
        mpv_z)))
except Exception as e:
    print(f"Edge enumeration failed: {e}")
    edge_pts = []
```

CadQuery



.jpg

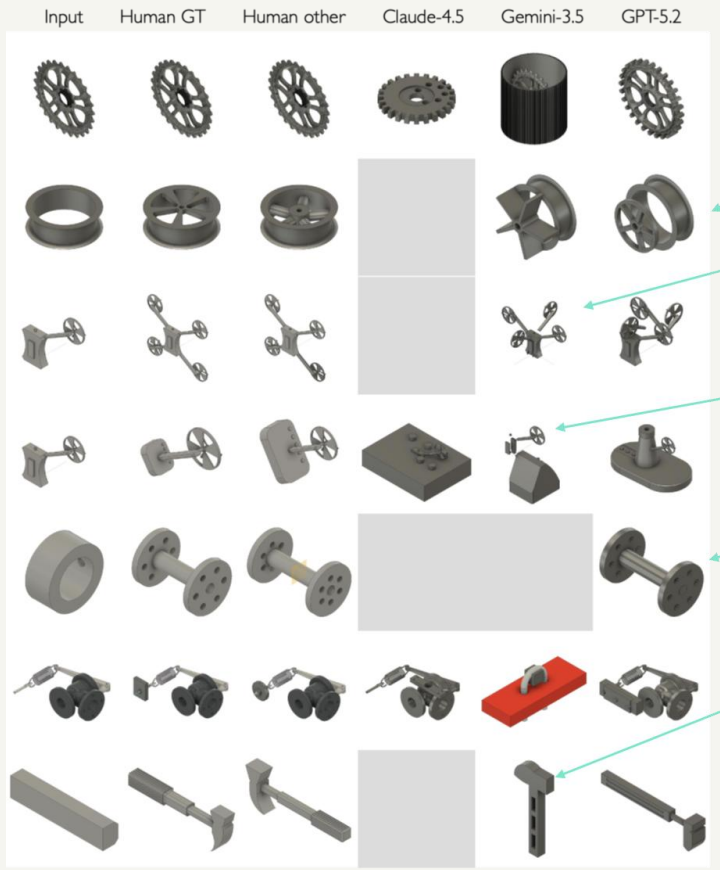


.step

Room for Improvement

Maximize performance on automatic metrics

- Pass additional metadata about the CAD model and its current state
- Use multiple agents to perform and guide different parts of the edit process
- Provide proving tools to the VLM
- Improve token efficiency
- Reduce the cost of performing edits



Creates spokes, misplaces them

Gets additional rotors, misplaces them

Roughly understands instruction, but just really bad

OK!

Can't edit existing model, so starts from scratch

AI currently lacks 3D visual understanding

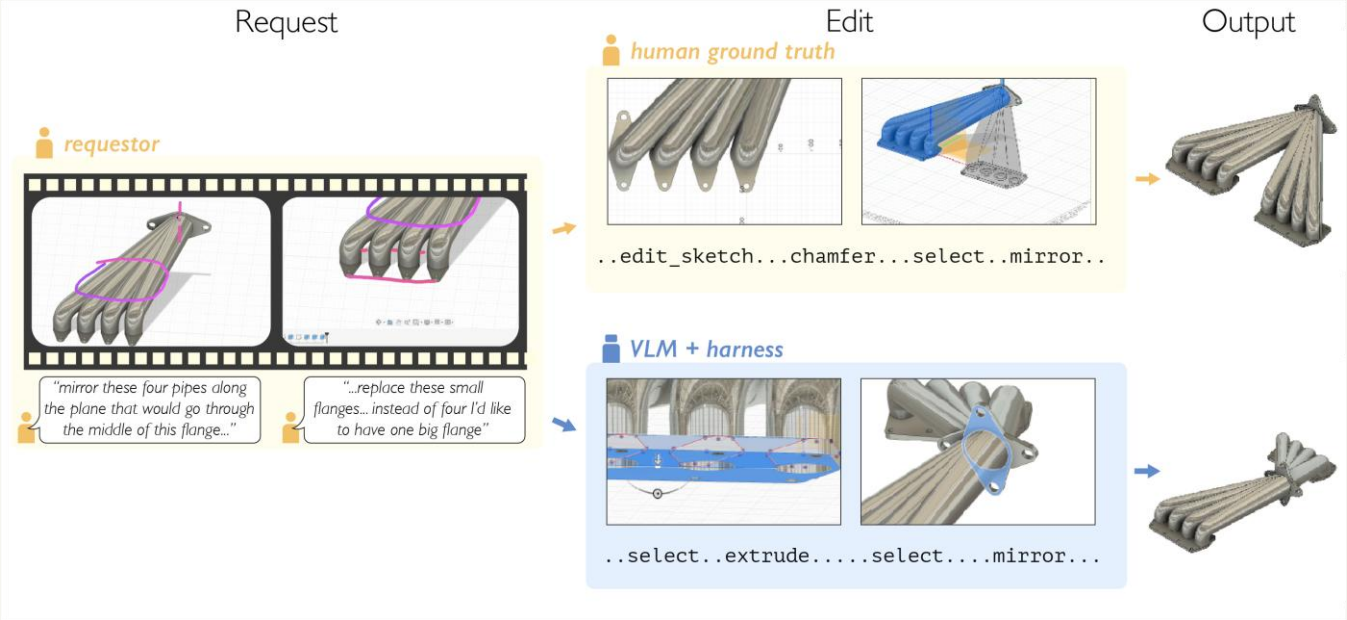
The Benchmark



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Benchmark

- 48 edit requests total
  - Text-conditioned edits only (not video/drawing)
  - Collected from expert designers
  - Some assemblies, some single components
  - Expected to take human designers between 2 and 10 minutes each
- Link for download in repo
- 3 automatic evaluation scripts



Submission Details

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Submission Details

- Code
  - Submit your final code as a pull request to our repository
- Output CAD files (.step and .stl)
  - Include in the repository
- Scores (computed by code we provide):
  - Surface chamfer similarity,
  - Volumetric F1 similarity,
  - Volumetric *Difference* F1 similarity <-- most important!
- Final presentation
  - Include a PDF of the presentation in your repository
  - Include overview of your method, final scores (metric_bar_facets.png), images of the 5 selected test examples for qualitative evaluation on 1 slide

Category	Criteria	Score
Quantitative Evaluation (40%)	On all text-conditioned edits (48), computed with provided codebase: • **Diff F1** • Volumetric F1 • Chamfer Similarity	0-10 points
Qualitative Evaluation (40%)	Method evaluation: • Novelty • Cost efficiency • Scalability • Effectiveness	0-10 points
Other (20%)	Presentation quality, structure, and clarity. Future work and broader impact discussion.	0-10 points

```
test_example_request_ids = [
'SUJ2G2UMJQR7PMBX_1759209987.785593',
'3YH2WFSRM22W7DKT_1769773335.525203',
'B7A2N74ZJBF9MZHU_1770174133.012106',
'F332D3FXML85WLR2_1769607142.566352',
'ZK22J6VYRKQ2RTFD_1758874422.1403751']
```

Good luck!

Get started with code and dataset > https://github.com/grndnl/IDETC26-Hackathon-Autodesk-neuralCAD-Edit



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